

Spatial Pressure Gravitational Theory — Public

Disclosure of Technical Anchors

Released under the Intellectual Property Spacetime Proof Protocol (IPSTP v1.0)

Prepared in accordance with IPSTP v2.0 Engineering Implementation and Legal Interpretation Specification

Layer A + Layer B Public Document

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The integrity of this public document itself is recorded and verifiable through the external platforms referenced above. The verification layer file is an encrypted archive of SGT's core technical assets. Its hash (spatial anchoring), timestamp (temporal anchoring), and snapshot DOI together constitute an immutable spatiotemporal evidence chain. The three elements cross-reference one another; the absence of any one renders the evidence chain incomplete.

Declaration

This document is the full technical anchor disclosure of the Spatial Pressure Gravitational Theory (SGT), prepared in accordance with the IPSTP v1.0 three-layer 剥离 structure, with the core principle being the separation of verification rights and access rights.

Layer A (Condition and Conclusion Layer) publicly discloses all theoretical conditions and precise numerical conclusions of SGT. Layer B (Constraint Framework Layer) publicly discloses all theoretical constraints and structural rules of SGT. The above information satisfies the IPSTP v1.0 Verification Sufficiency Principle — it is sufficient to enable researchers with relevant numerical backgrounds to independently verify the conclusions through independent engineering pathways,

without any need to access Layer C.

Layer C (Implementation Source Layer) is retained as a trade secret in accordance with IPSTP v1.0. Layer C shall be directionally decrypted under compliant procedures only in legally prescribed circumstances, namely judicial evidence collection, administrative investigation, or written authorization by the author.

Routine verification is absolutely exempt.

This document constitutes a Verifiable Disclosure Event (VDE). Should identical or highly similar condition-conclusion mappings subsequently appear, independent verification may be conducted through the spatiotemporal anchoring records established herein.

(Note: In the anchor numbering below, P = Premise (conditions), B = Boundary (constraints), Y = Yield (conclusions), belonging respectively to Layer A and Layer B of IPSTP.)

I. Theoretical Project

Theory Name: Spatial Pressure Gravitational Theory (SGT)

Theoretical Positioning: An effective gravitational theory with a continuous elastic medium as its physical ontology. Matter constitutes localized defects within the medium; gravity is the tension gradient generated when the medium is distended by matter. The theory covers all gravitational scenarios from galactic rotation curves to cosmological big bang.

Theoretical Domains: Foundational Theory | Lagrangian Field Theory | Spherically

Symmetric Black Holes | Rotating Black Holes | Primordial Universe and Singularity
 Resolution | Cosmological Linear Perturbations and CMB | Compact Binary
 Dynamics | Gravitational Structural Stability | Testable Observational Predictions

II. Premise Set P (Belonging to Layer A)

2.1 Axiom System

Number	Axiom	Content
SGT-P001	A1	The spatial pressure medium exists. The vacuum is filled by a continuous elastic medium.
SGT-P002	A2	Matter is a localized defect in the medium, distending the surrounding medium when occupying space.
SGT-P003	A3	Gravity is the tension gradient of the medium. A tension gradient produces a net force directed toward the center.

All conclusions are derived from the above three axioms.

2.2 Core Variables and Field Equations

Number	Condition	Content
SGT-P004	Distension degree	$f = 1 - A_0 \in [0,1]$
SGT-P005	Order parameter constitutive law	$\chi = (1-f)/(1+f^2), \chi \in [0,1]$

SGT-P006	Effective gravitational constant	$G_{\text{eff}} = G \cdot \chi(f)$
SGT-P007	Static field equation form	Reduces to the Ω -A1 ODE system under spherical vacuum symmetry
SGT-P008	Modified Friedmann equation	$H^2 = (8\pi G/3)\chi\rho_m + (8\pi GK/3)\rho_T(f)$, derived from variational action
SGT-P009	f dynamics equation	$\ddot{f} + 3H\dot{f} + V_{\text{eff}}(f) = 0$, derived from variational action

2.3 Parameter Statement

Number	Condition	Content
SGT-P010	Number of independent parameters	2 — the medium coupling constant K and the critical distension degree f_c . Δf is determined by f_c and is a non-independent parameter.

2.4 Lagrangian Field Theory Framework

Number	Condition	Content
SGT-P011	Total action structure	$S = S_{\text{medium}} + S_{\text{matter}}$, containing the standard Einstein-Hilbert term and an elastic strain energy term
SGT-P012	Kinetic energy term	Under Lagrangian description, the kinetic energy term $F=1$ is exact (Grade A closure)
SGT-P013	Stress constitutive relation	Derived from the first variation of the elastic strain energy, fully explicit and closed

III. Constraint Set B (Belonging to Layer B)

3.1 Non-emptiability and Singularity Resolution

Number	Constraint	Strength of Proof
SGT-B001	$\chi \geq 0$ always holds; the medium cannot be completely expelled	Grade A multi-dimensional proof
SGT-B002	Equivalence proposition: $C_{rr} \rightarrow \infty \iff \chi \geq 0$	Grade A
SGT-B003	$f \leq 1$ locked at the action level	Grade A
SGT-B004	Cosmic censorship hypothesis: non-emptiability prohibits naked singularities	Theorem-grade corollary

3.2 Elastic Constant System

Number	Constraint	Strength of Proof
SGT-B005	Physical state of the medium: prestressed transversely isotropic elastic solid	Grade A
SGT-B006	Anisotropic stress fingerprint: $P_\theta < P_r$ holds globally	Grade A
SGT-B007	Elastic constant system closure: 6 constants (4 independent + 2 constrained)	Grade A-
SGT-B008	As $\chi \rightarrow 0$: $C_{rr} \rightarrow \infty$, $C_{\theta\theta} \rightarrow 0$, $C_{44} \rightarrow 0$, $C_{66} \rightarrow 0$	Grade A- cross-validation

SGT-B009	χ as order parameter: organizing elastic phase transitions	Grade A
SGT-B010	Frozen layer structure: five layers, freezing front $\approx$ photon sphere	Grade B+
SGT-B011	No solitary wave solutions exist	Grade A

3.3 Causal Structure and Stability

Number	Constraint	Strength of Proof
SGT-B012	Causal structure theorem: inner shell is causally closed	Grade A
SGT-B013	Global hyperbolicity	Grade A
SGT-B014	Energy conservation: $dE/dt=0$	Grade A
SGT-B015	Nonlinear stability rigorously proven	Grade A
SGT-B016	Onsager reciprocity holds exactly	Grade A
SGT-B017	Static solution uniqueness theorem	Grade A
SGT-B018	Qualitative equivalence theorem for $\chi(f)$	Grade A-
SGT-	Metric-medium separation principle	Rigorously

B019		proven
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3.4 Black Hole Physics

Number	Constraint	Strength of Proof
SGT-B020	Curvature scalars finite at the absolute horizon	Grade A
SGT-B021	Smarr formula holds exactly	Grade A
SGT-B022	Area law holds strictly	Grade A

3.5 Rotating Black Holes

Number	Constraint	Strength of Proof
SGT-B023	Local Effective Mass Approximation validated for $a \leq 0.5$	Grade A
SGT-B024	Geff negative feedback mechanism	Grade A

3.6 Cosmology

Number	Constraint	Strength of Proof
SGT-B025	Singularity resolution theorem: $f(0)=1 \rightarrow H(0)<\infty$	Grade A
SGT-	Equivalence theorem: $f(0)=1 \iff \chi(0)=0 \iff H(0)<\infty$	Grade A

B026		
SGT-B027	Primordial phase structure: frozen de Sitter $\rightarrow$ thawing $\rightarrow$ fast-roll $\rightarrow$ normal expansion	Qualitatively closed
SGT-B028	Unified mechanism with black hole singularity resolution	Grade A
SGT-B029	Standard slow-roll inflation does not hold (honest conclusion)	Noted
SGT-B030	Scalar perturbation suppression in frozen phase: $\chi=0 \rightarrow G_{\text{eff}}=0$	Grade A

3.7 Compact Binaries

Number	Constraint	Strength of Proof
SGT-B031	χ as intrinsic scalar degree of freedom	Grade A
SGT-B032	Scalar dipole radiation originates from χ temporal variation	Qualitative

3.8 Gravitational Structural Stability

Number	Constraint	Strength of Proof
SGT-B033	Three-tier rigidity map: horizon ε^0 , ISCO ε^2 , photon sphere ε^1	Grade A
SGT-	Safety margin: $71\times$	Grade A

B034		
SGT-B035	All 8 CH-series tasks completed	Grade A

IV. Conclusion Set Y (Belonging to Layer A)

4.1 Spherically Symmetric Black Holes

Number	Conclusion	SGT Value	GR Value	Strength of Proof
SGT-Y001	ISCO radius	4.23M	6.00M	Grade A
SGT-Y002	Horizon radius	1.6673M	2M	Grade A
SGT-Y003	Photon sphere radius	2.9954M	3M	Grade A
SGT-Y004	Critical impact parameter	5.1953M	5.1962M	Grade A
SGT-Y005	Hawking temperature ratio	$0.866 \times T_H^{\text{GR}}$	T_H^{GR}	Grade A
SGT-Y006	Horizon entropy decomposition	$\eta = 1.66 = 1.00 + 0.66$	1.00	Grade B

4.2 Rotating Black Holes

Number	Conclusion	SGT Value	GR Value	Strength of Proof
SGT-Y007	ISCO rotation coefficient C1	-0.466	-3.266 (86% suppression)	Grade A
SGT-Y008	Observable region difference (slow rotation)	<0.3%	—	Grade A
SGT-Y009	Shadow D-shape difference	<0.05%	—	Grade A
SGT-Y010	ISCO-Kerr crossing point	$a \approx 0.5$	—	Grade A
SGT-Y011	Quadrupole moment deviation	$Q/Q_{\text{Kerr}} \approx 0.58$	—	Grade B

4.3 Primordial Universe

Number	Conclusion	Value	Strength of Proof
SGT-Y012	Initial Hubble constant $H(0)$	1.023×10^{-1}	Grade A
SGT-Y013	Initial total density	1.250×10^{-3}	Grade A

4.4 CMB Fingerprints

Number	Conclusion	Characteristic	Strength of Proof
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SGT-Y014	Low- ℓ suppression of scalar spectrum	$\chi \rightarrow 0 \rightarrow G_{\text{eff}} \rightarrow 0$	Grade A
SGT-Y015	Early Ψ - Φ separation	Elastic anisotropy	Grade A
SGT-Y016	Super-horizon $\mathcal{R}$ non-conservation	$\delta P_{\text{Nad}} \neq 0$	Grade A
SGT-Y017	Low- ℓ enhancement of tensor-to-scalar ratio r	P_{T} normal, P_{S} suppressed	Qualitative
SGT-Y018	Tachyon perturbation generation	Driven by $V_{\text{eff}}' < 0$	Grade B+

4.5 Gravitational Structural Stability

Number	Conclusion	Characteristic	Strength of Proof
SGT-Y019	ISCO metric perturbation response	ε^2 -order, first-order strictly zero	Grade A
SGT-Y020	Photon sphere metric perturbation response	ε^1 -order	Grade A
SGT-Y021	Horizon metric perturbation response	ε^0 -order, invariant	Grade A

4.6 Testable Predictions

Number	Prediction	Value/Feature	Verification Instrument	Strength of Proof
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SGT-Y022	Fe K α line red-wing extension	500-900 eV	Athena 2027+	Grade A
SGT-Y023	Gravitational wave echo time delay	$\sim 14.3\text{M}$	3G detectors	Grade A
SGT-Y024	Echo frequency dependence	$R(\omega) \propto \omega^\alpha$	3G detectors	Grade A
SGT-Y025	Neutron star TOV — no modification	—	NICER verified	Grade A
SGT-Y026	Rotating black hole shadow	Deviation $< 0.3\%$	EHT verified	Grade A
SGT-Y027	CMB scalar spectrum low- ℓ suppression	$\ell < 30$	CMB-S4	Grade A
SGT-Y028	CMB early Ψ - Φ separation	E-mode	CMB-S4	Grade A
SGT-Y029	ISCO rotation suppression	$C_1 = -0.466$	LISA 2035+	Grade A
SGT-Y030	Primordial GW low-frequency suppression	$G_{\text{eff}} \rightarrow 0$	3G detectors	Qualitative

V. Layer C Retention Statement

The core technical assets of the SGT (derivation algorithms, solution codes, and other engineering implementation details) are retained as Layer C (Implementation Source

Layer) in accordance with IPSTP v1.0. They are permanently and privately preserved by default, with trade secret protection applied throughout.

Layers A and B have already provided sufficient information for third parties to reconstruct constraints and independently verify conclusions. Layer C is absolutely not a necessary condition for verification.

Legal Access Exceptions: Layer C shall be directionally decrypted under compliant procedures only in legally prescribed circumstances — judicial evidence collection, administrative investigation, or written authorization by the author — serving as the ultimate fallback asset for intellectual property rights.

VI. Intellectual Property Protection System

Statement

The intellectual property protection of the Spatial Pressure Gravitational Theory has established a multi-layered, cross-platform, judicially-grade deposit and verification system, realizing the separation of verification rights and access rights:

Affirmation Protocol Layer: IPSTP v1.0 Intellectual Property Spacetime Proof Protocol — transforms theoretical publication into Verifiable Disclosure Events (VDE) through the three-layer 剥离 structure and spatiotemporal dual-anchoring.

Spatiotemporal Anchoring Layer: SHA-256 hash 固化 (spatial anchoring) + TSA trusted timestamp/blockchain deposit (temporal anchoring) — multi-source cross-verified, immutable spatiotemporal evidence.

Academic Deposit Layer: Zenodo DOI permanent archive, academic preprint DOI registration — globally searchable records of academic priority.

Legal Filing Layer: National Copyright Center registration, DCI Digital Copyright Identifier — national affirmation with legal presumption of rights.

Technical Verification Layer: ZKAV Zero-Knowledge Academic Verification Protocol — undertakes closed-book solution verification of external physical propositions without disclosing Layer C.

The above systems are mutually independent and take effect synergistically. The intellectual property protection of SGT does not rely on any single institution or jurisdiction, but has established independently admissible multiple evidence structures within mainstream global judicial jurisdictions. Beyond the mechanisms enumerated above, SGT has deployed no fewer than 10 independent, forensically traceable deposit mechanisms, forming a three-dimensional, full-chain evidence coverage. One illustrative example: regularly scheduled email deposit filings to over 100 industry media outlets, academic institutions, and domain experts — creating a large-scale distributed priority witness network through independent third-party communication records. The timestamp, transmission/receipt logs, and content archive of each filing email constitute an independent, forensically traceable piece of evidence.

VII. Directional Statement

The 78 technical anchors disclosed in this document — 13 premises P, 35 constraints

B, 30 conclusions Y — collectively constitute the non-detachable theoretical fingerprint of the Spatial Pressure Gravitational Theory. The identification of this fingerprint is independent of the mathematical representation form or derivation path of the theory, being determined solely by the numerical and structural consistency of the anchors.

When any subsequent theory simultaneously satisfies the following circumstances, it shall be deemed to fall within the condition space already established by SGT:

- (a) Its axiom system is substantially similar to SGT-P001 through SGT-P003 of this document;
- (b) Its field equation framework and constitutive law form are equivalent expressions of SGT-P004 through SGT-P009 of this document;
- (c) Its numerical conclusions exhibit systematic consistency with SGT-Y001 through SGT-Y030 of this document.

When the above circumstances are satisfied, that theory does not constitute an independent discovery, but rather a re-implementation of the condition-conclusion mapping already publicly disclosed by SGT. The spatiotemporal anchoring records established by this document through IPSTP v1.0 Tier 3 preservation and the aforementioned multiple judicial deposit systems constitute a complete electronic evidence chain for claiming first disclosure. Pursuant to the IPSTP v1.0 protocol, should a later party claim independent discovery, they must prove either that they independently completed the establishment of the same mapping prior to the archival date of this document, or that their condition system bears a substantive, incommensurable difference from this document.

Any unauthorized appropriation will face judicial scrutiny with multi-source spatiotemporal evidence cross-verification. The publication of this document has rendered the risk-to-reward ratio of plagiarizing SGT's core achievements severely imbalanced — the traceability, provability, and legal accountability of infringing acts have been pre-emptively locked at the technical level.

VIII. Mandatory Standard Annotation

This paper follows the IPSTP v1.0 Tier 3 preservation mechanism. Spatial anchoring and temporal anchoring have been completed. Archive number: 10.5281/zenodo.20558741. Timestamp number: TSA-01-20260605694921944. Layer C is retained as a trade secret.

IX. Anchor Statistics

Category	Count
Premises P (Layer A)	13
Constraints B (Layer B)	35
Conclusions Y (Layer A)	30
Total Anchors	78
Theoretical Domains	9

IPSTP Preservation Tier	Tier 3 (Maximum Preservation · Judicial-Grade Hardened)
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Spatial Pressure Gravitational Theory — Public Disclosure of Technical Anchors.

IPSTP v1.0 Tier 3 Preservation. Jointly Secured by Spatiotemporal Dual-Anchor.